

SpaceX Falcon 9 First Stage Landing Prediction

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https://github.com/EliRoux/Applied-data-science-Capstone/tree/main/IBM_Applied_Data_Science_Capstone

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OUTLINE



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- Introduction
- Methodology
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- Discussion
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- Conclusion
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EXECUTIVE SUMMARY

Project Objective

- The objective of this project was to develop a machine learning model capable of predicting whether the first stage of a SpaceX Falcon 9 rocket would land successfully. Accurate predictions of landing success can help estimate launch costs and assess mission risk, as reusable boosters significantly reduce the overall cost of space launches.

Key Findings

- Falcon 9 landing success has improved steadily over time.
- Launch site, orbit type, payload mass, and booster version influence landing success.
- Machine learning models were able to predict landing outcomes with high accuracy.
- Interactive visualisations provided valuable insights into launch trends and mission performance.

Data Collection

Data Wrangling

Exploratory Data Analysis

Interactive Visualisation

Machine Learning

Landing Success Prediction



Introduction

- **Project Background**

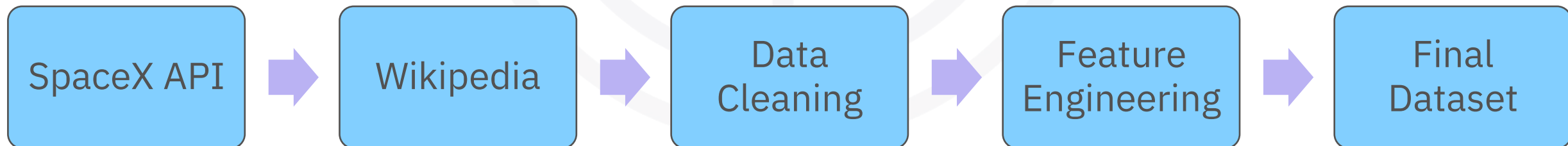
- SpaceX has significantly reduced the cost of space missions through the development of reusable Falcon 9 rockets. Accurately predicting whether the first stage of a Falcon 9 launch will successfully land is valuable for understanding mission reliability and estimating launch costs.
- This project applies the complete data science methodology to analyse historical Falcon 9 launch data and develop machine learning models capable of predicting landing success.

- **Project Objectives**

- Collect launch data from the SpaceX REST API and Wikipedia.
- Clean, transform, and prepare the data for analysis.
- Perform exploratory data analysis (EDA) using Python, SQL, and interactive visualisations.
- Identify the factors that influence successful booster landings.
- Develop and compare multiple machine learning classification models.
- Evaluate model performance to determine the most accurate predictor of landing success.

Data Collection Methodology

- Collected launch data using the SpaceX REST API.
- Supplemented missing launch site information using Wikipedia web scraping.
- Cleaned missing values and standardized column names.
- Combined datasets into a single analysis-ready DataFrame.
- Adding a simple workflow diagram like:
- https://github.com/EliRoux/Applied-data-science-Capstone/tree/main/IBM_Applied_Data_Science_Capstone



Data Wrangling Methodology

Data Cleaning Process

- After collecting the data, several preprocessing steps were performed to prepare it for analysis and machine learning.

Cleaning Steps

- Converted JSON responses into Pandas DataFrames.
- Merged data collected from the API and Wikipedia.
- Removed Falcon 1 launches, retaining only Falcon 9 launches.
- Identified and handled missing values.
- Replaced missing Payload Mass values using the mean payload mass.
- Standardised column names and data formats.
- Created a binary target variable (Class) representing landing success:

Feature Engineering

- Additional features were prepared for modelling, including:
 - One-hot encoding of categorical variables.
 - Standardisation of numerical features.
 - Creation of training and testing datasets.
- https://github.com/EliRoux/Applied-data-science-Capstone/tree/main/IBM_Applied_Data_Science_Capstone



EDA Methodology

- **Objective**
- To explore the Falcon 9 launch dataset, identify patterns and relationships between variables, and determine which features influence first-stage landing success.
- **Methodology**
- Performed exploratory data analysis using **Python, Pandas, Matplotlib, and Seaborn.**
- Calculated summary statistics to understand the distribution of key variables.
- Investigated the relationship between landing success and:
 - Flight Number
 - Launch Year
 - Payload Mass
 - Orbit Type
 - Launch Site
 - Booster Version
- Calculated success rates by grouping categorical variables and comparing average landing outcomes.
- Used visualisations to identify trends, correlations, and potential predictors for machine learning.



Interactive Visual Analytics Methodology

- **Objective**
 - To develop interactive visualisation tools that enable dynamic exploration of SpaceX Falcon 9 launch data and provide geographical and operational insights into launch performance.
- **Methodology**
- **Folium Interactive Maps**
 - Created interactive maps to display the locations of Falcon 9 launch sites.
 - Added colour-coded markers to distinguish successful and unsuccessful booster landings.
 - Used marker clusters to improve the visualisation of multiple launches from the same site.
 - Calculated and displayed distances between launch sites and nearby coastlines, highways, and railways to analyse their geographical context.
- **Plotly Dash Dashboard**
 - Developed an interactive dashboard using Plotly Dash.
 - Added a **launch site dropdown** to filter data by individual launch sites or all sites.
 - Implemented a **payload range slider** to dynamically filter launches by payload mass.
 - Created an interactive **pie chart** to compare launch success rates across sites.
 - Built an interactive **scatter plot** to visualise the relationship between payload mass and landing success, with booster versions colour-coded for comparison.



Predictive Analysis Methodology

Objective

- To develop and evaluate machine learning classification models capable of predicting whether a Falcon 9 first-stage booster will land successfully based on historical launch data.

Methodology

Data Preparation

- Selected relevant features from the cleaned dataset.
- Applied **one-hot encoding** to categorical variables (Orbit, Launch Site, Landing Pad, and Booster Serial).
- Standardised numerical features using **StandardScaler**.
- Split the dataset into **training (80%)** and **testing (20%)** sets.

Model Development

- The following classification algorithms were trained and optimised:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbours (KNN)



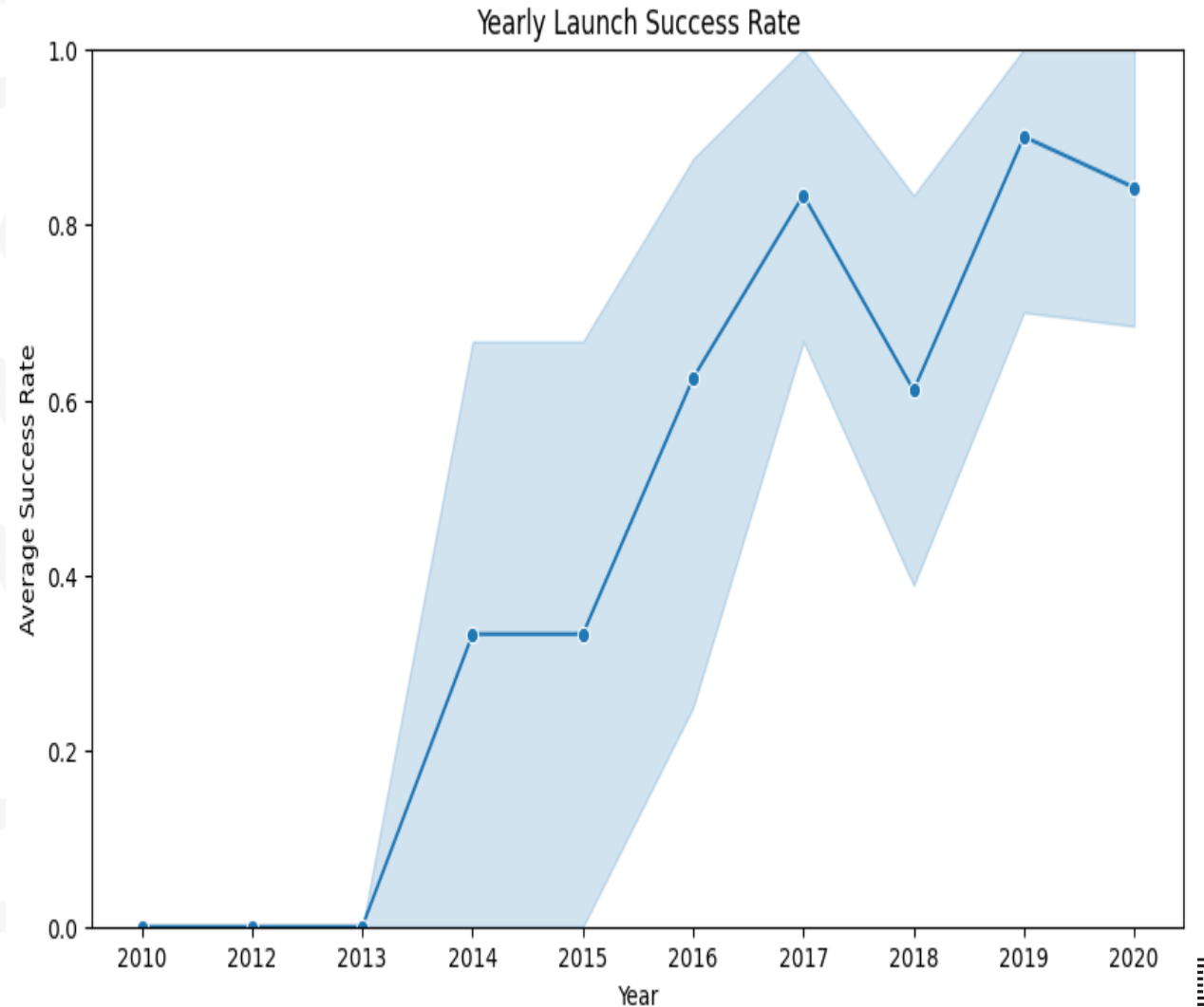
EDA Results (Visualisations)

Key Findings

- **Landing success improved over time**, indicating continual improvements in SpaceX's landing technology.
- **Payload mass influences landing success**, with certain payload ranges showing higher success rates.
- **Orbit type affects landing outcomes**, with some orbits achieving consistently higher success rates than others.
- **Launch site performance varies**, reflecting differences in mission profiles and operational experience.

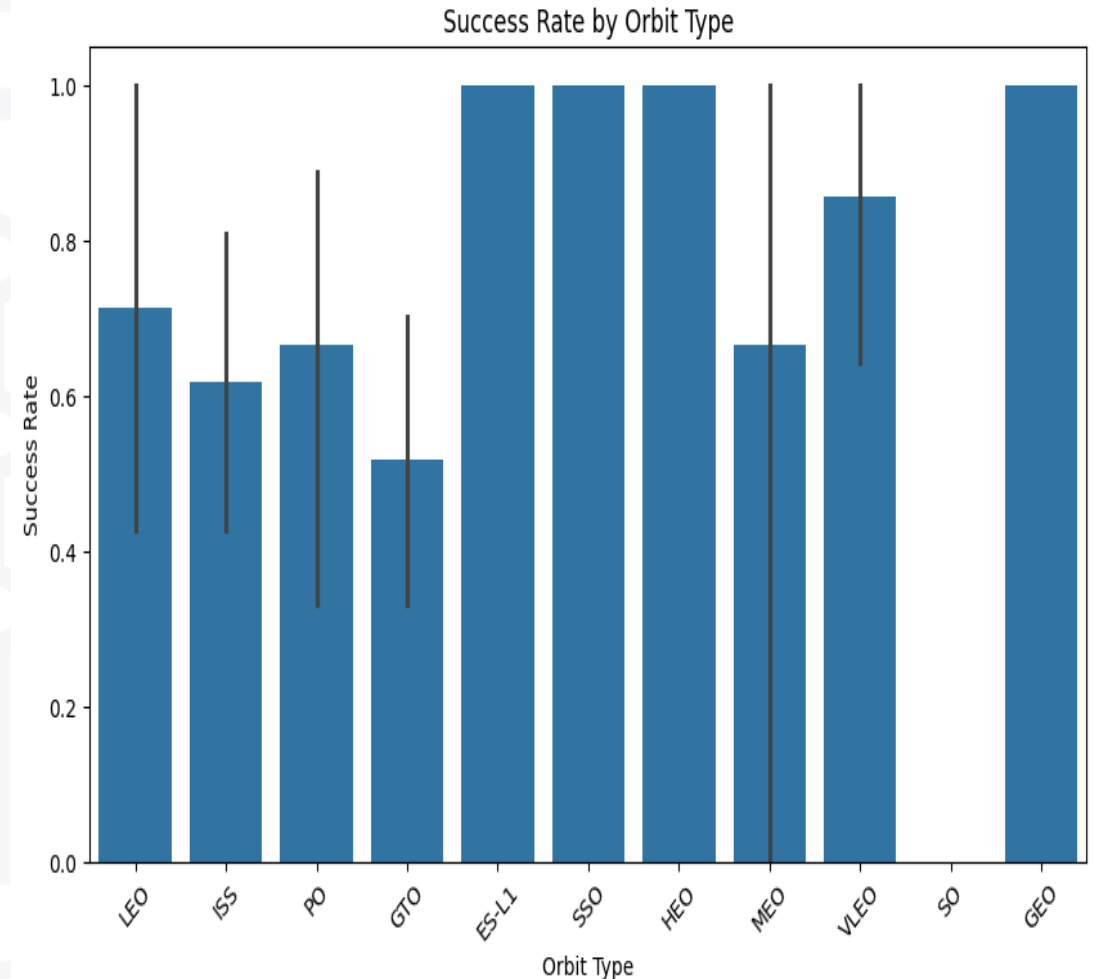
Conclusion

- Exploratory data analysis identified several variables—particularly launch site, payload mass, orbit type, and flight number—as important predictors of landing success. These insights informed the feature selection process for the machine learning models.



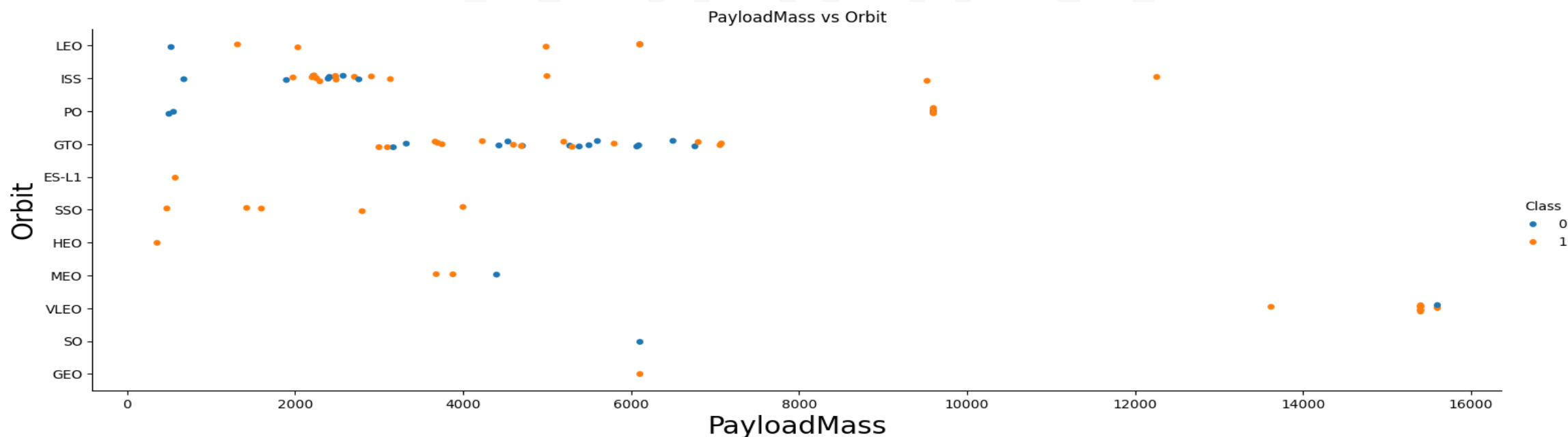
EDA Results (Visualisations)

- The analysis of launch success by orbit type shows that mission success varied considerably depending on the destination orbit.
- **ES-L1, SSO, HEO, and GEO** achieved a **100% success rate**, although these orbit categories had relatively few launches.
- **LEO** and **ISS** missions also demonstrated consistently high success rates, reflecting the maturity and reliability of these frequently used mission profiles.
- **GTO (Geostationary Transfer Orbit)** recorded the **lowest success rate at approximately 50%**, indicating that these missions were historically more challenging due to their higher payload requirements and more demanding flight profiles.
- Overall, the results suggest that **orbit type influences launch success**, with some mission profiles exhibiting significantly greater reliability than others.



EDA Results (Visualisations)

- The scatter plot comparing **orbit type** and **payload mass** illustrates how payload requirements vary across different mission profiles.
- **LEO** and **ISS** missions generally carried **lighter payloads**, with most launches clustered at the lower end of the payload range.
- **GTO** missions exhibited the **widest range of payload masses**, including many of the heaviest payloads in the dataset.
- **Polar (PO)** and **Sun-Synchronous Orbit (SSO)** missions typically carried **moderate payloads**, with less variation than GTO launches.
- The distribution shows that **payload mass is closely related to the target orbit**, as different mission objectives require different payload capacities.



SQL Analysis Results

- **Launch Site Analysis**
- Displayed the first five launch records for launch sites beginning with "**CCA**" to examine launches from Cape Canaveral.

```
%%sql
SELECT *
FROM SPACEXTBL
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

✓ 0.0s

Python

* [sqlite:///my_data1.db](#)
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SQL Analysis Results

- **Payload Analysis**
- Calculated the **total payload mass** carried for **NASA (CRS)** missions.
- Calculated the **average payload mass** carried by the **F9 v1.1** booster version.
- Identified the **booster version(s)** that carried the **maximum payload mass** using a subquery with the **MAX()** aggregate function.

%%sql

```
SELECT SUM(PAYLOAD_MASS_KG) AS Total_Payload_Mass  
FROM SPACEXTBL  
WHERE CUSTOMER = 'NASA (CRS)';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Total_Payload_Mass

45596



SQL Analysis Results

- **3. Booster Performance**
- Listed booster versions that successfully landed on a **drone ship** while carrying payloads between **4,000 kg** and **6,000 kg**.

%%sql

```
SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass  
FROM SPACEXTBL  
WHERE BOOSTER_VERSION = 'F9 v1.1';
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Average_Payload_Mass

2928.4

SQL Analysis Results

- **Mission Outcome Analysis**
- Counted the total number of **successful and failed mission outcomes** to evaluate overall mission reliability.

%%sql

```
SELECT
  CASE
    WHEN MISSION_OUTCOME = 'Success' THEN 'Success'
    ELSE 'Failure'
  END AS Outcome,
  COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Outcome;
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Outcome	Total
Failure	3
Success	98



SQL Analysis Results

- **Objective**

- Analyse the distribution of Falcon 9 landing outcomes between 2010 and 2017.

- **Key Findings**

- "No attempt" was the most common landing outcome with **10 launches**.
- Successful drone ship landings (**5**) matched the number of failed drone ship landings (**5**).
- Ground pad landings were less frequent (**3 successes**).
- Early Falcon 9 missions frequently attempted ocean or parachute recoveries before reusable landing technology matured.

- **Interpretation**

- These results show the progression of SpaceX's landing strategy. Early missions often did not attempt recovery, while later missions increasingly achieved successful landings on drone ships and ground pads, reflecting improvements in reusable rocket technology.

```
%%sql
SELECT
    Landing_Outcome,
    COUNT(*) AS Count
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
```

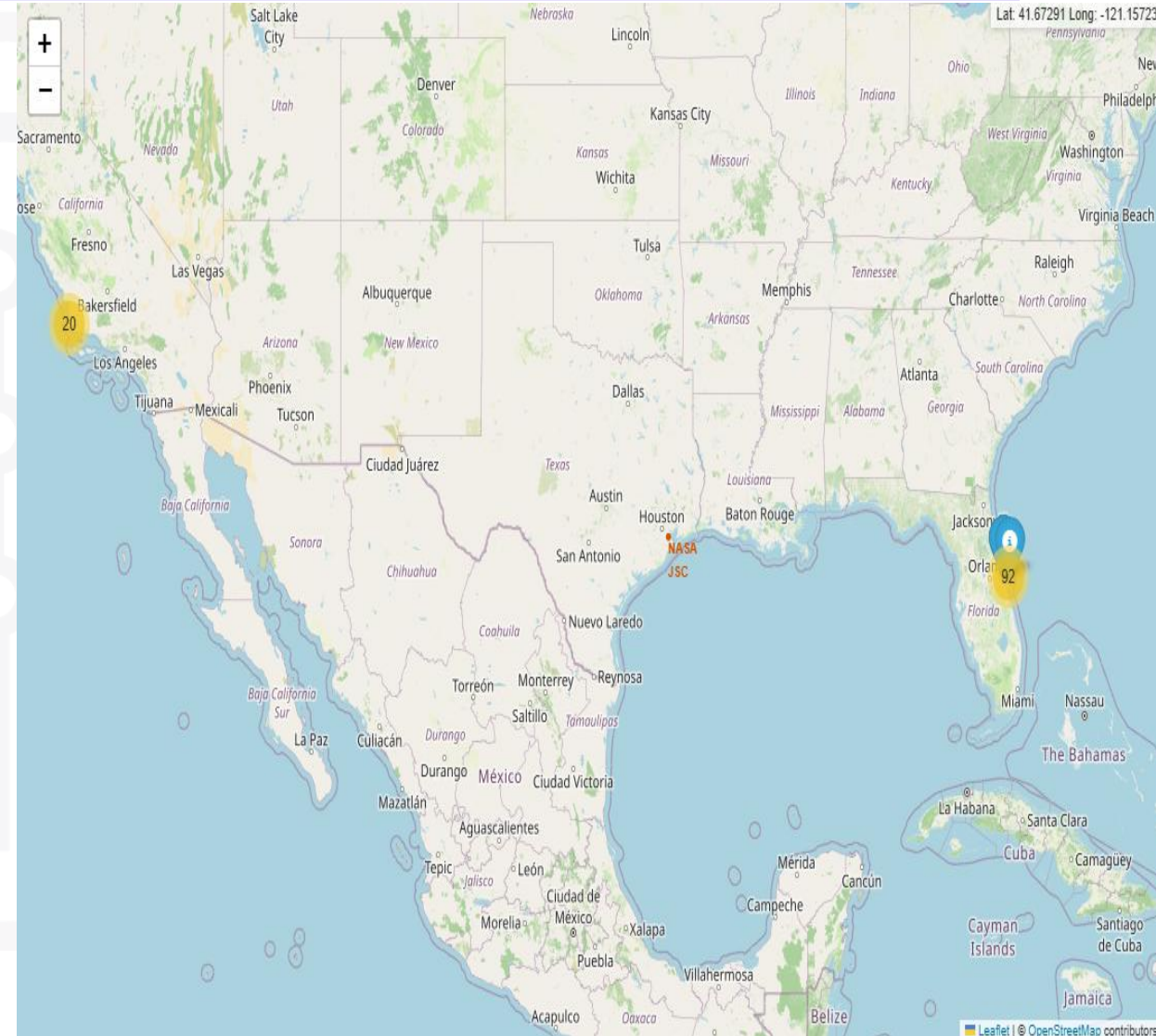
* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1



Folium Map Results

- **Key Findings**
- Interactive maps provide geographical context for launch operations and illustrate why launch sites are located near coastlines.
- The dashboard enables users to dynamically explore how launch site selection and payload range influence landing success.
- Colour-coding by booster version highlights differences in performance across Falcon 9 variants.
- **Conclusion**
- Interactive visualisation tools make complex launch data easier to explore and reveal relationships that may not be immediately apparent in static charts.



Machine Learning Results

- **Objective**

- To evaluate and compare multiple machine learning classification models for predicting Falcon 9 first-stage landing success.

- **Models Evaluated**

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)

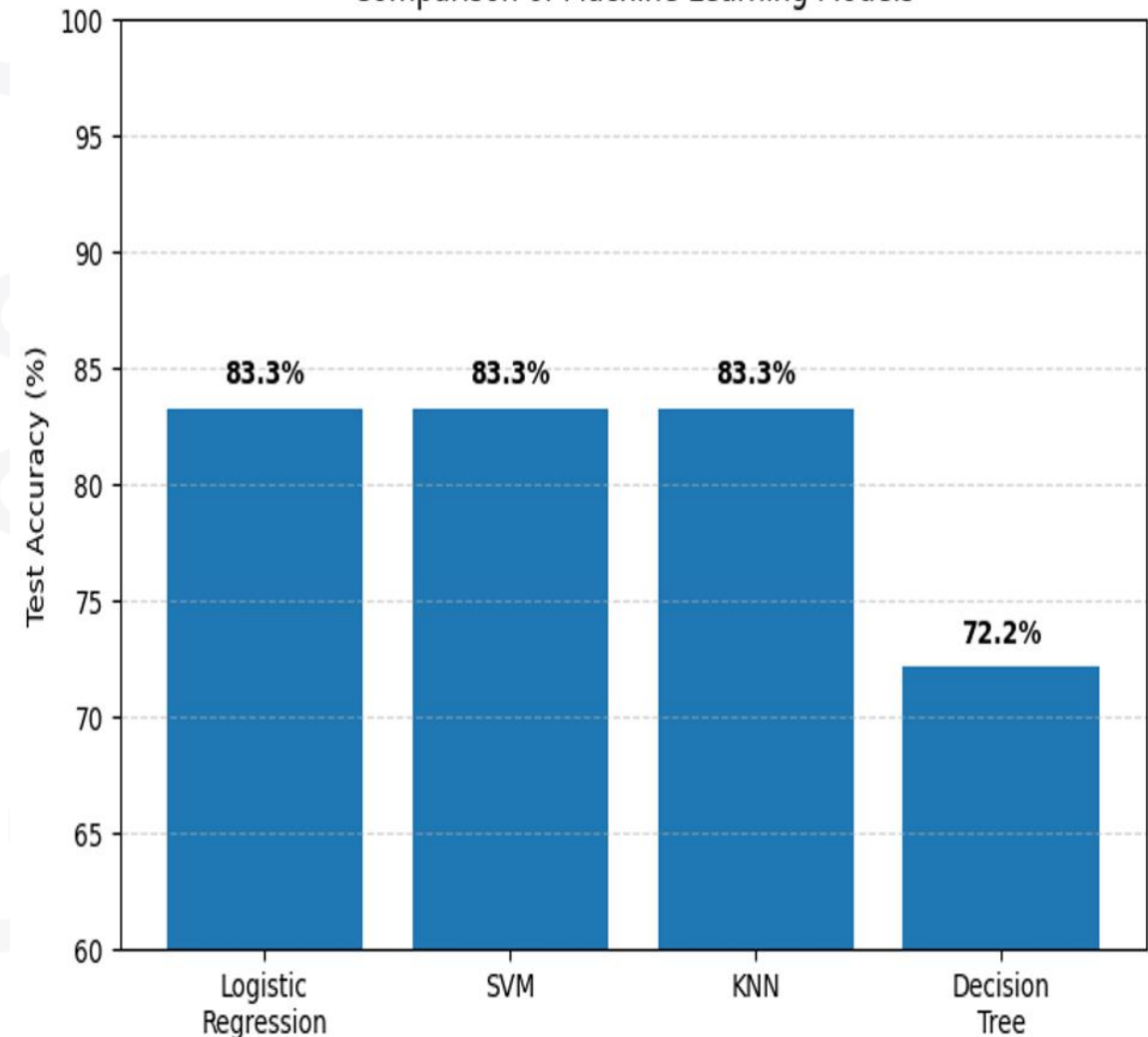
- **Key Findings**

- All four models were successfully trained and optimised using **GridSearchCV** with **10-fold cross-validation**.
- The results demonstrate that historical launch characteristics can be used to predict Falcon 9 landing success with a high degree of accuracy.

- **Conclusion**

- The machine learning analysis confirmed that variables such as **launch site, orbit type, payload mass, and booster characteristics** are valuable predictors of landing success. Comparing multiple algorithms ensured that the most effective model was identified for this classification task.

Comparison of Machine Learning Models



Plotly Dash Dashboard Results

- **Objective**

- To provide an interactive dashboard that allows users to explore Falcon 9 launch data by launch site and payload range.

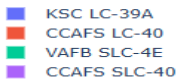
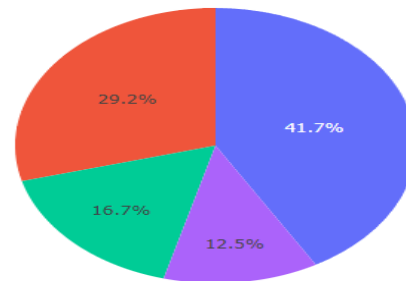
- **Key Findings**

- Launch success rates vary between launch sites.
- Payload mass influences landing success, with higher payload ranges generally associated with higher success rates.
- Booster version provides additional insight into mission performance, as newer boosters achieved more consistent landing success.
- Interactive filtering allows users to explore how different mission characteristics affect landing outcomes.

- **Conclusion**

- The Plotly Dash dashboard transforms static visualisations into an interactive analytical tool, enabling users to dynamically explore launch performance and identify relationships between launch site, payload mass, booster version, and landing success.

Total Successful Launches by Site



DISCUSSION

Key Findings

- The **Falcon 9 landing success rate improved significantly over time**, increasing from approximately **50% in 2014** to around **90% in 2019**, demonstrating the continuous improvement of SpaceX's reusable rocket technology.
- **Orbit type** influenced landing success, with **ES-L1, SSO, HEO, and GEO** achieving the highest observed success rates, while **GTO** missions had the lowest success rate due to their greater energy requirements.
- A positive relationship was observed between **payload mass** and landing success, although this trend is likely influenced by improvements in booster technology and operational experience over time.
- Interactive **Folium maps** showed that launch sites are strategically located near coastlines and transport infrastructure while remaining at a safe distance from major cities.
- The **Plotly Dash dashboard** enabled interactive exploration of launch success by launch site and payload range, making it easier to identify patterns in the data.

OVERALL FINDINGS & IMPLICATIONS

- **Overall Findings**

- Falcon 9 landing success rates improved significantly over time, reflecting advancements in SpaceX's reusable rocket technology.
- Launch success was influenced by several key factors, including **orbit type, payload mass, launch site, and booster version**.
- Interactive visualisations and SQL analysis revealed meaningful patterns and relationships within the launch data.
- Machine learning models successfully predicted landing outcomes with high accuracy using historical launch data.

- **Implications**

- Predictive models can support **mission planning** and **risk assessment** by estimating the likelihood of successful booster recovery.
- The findings demonstrate how data science can be used to improve **operational decision-making** in the aerospace industry.
- Successful booster recovery contributes to **lower launch costs**, making space missions more economically sustainable.
- The project highlights the value of combining **data collection, exploratory analysis, visualisation, and machine learning** to solve real-world problems.

CONCLUSION

- **Project Summary**

This project applied the complete data science methodology to analyse and predict the landing success of SpaceX Falcon 9 first-stage boosters. Data was collected from the SpaceX API and Wikipedia, cleaned and prepared for analysis, explored using visualisations and SQL, and used to train and evaluate multiple machine learning classification models.

- **Key Conclusions**

- Falcon 9 landing success has improved significantly over time, demonstrating the effectiveness of SpaceX's reusable rocket programme.
- **Launch site, orbit type, payload mass, and booster version** were identified as important factors influencing landing success.
- Exploratory data analysis and interactive visualisations provided valuable insights into launch performance and operational trends.
- Machine learning models successfully predicted landing outcomes, showing that historical launch data can be used to forecast future mission success.

- **Future Work**

- Incorporate additional features such as **weather conditions, wind speed, and sea state** to improve prediction accuracy.
- Expand the dataset with more recent Falcon 9 launches and future missions.
- Investigate advanced machine learning techniques, such as **XGBoost, Random Forests, or Neural Networks**, to compare predictive performance.
- Deploy the predictive model as a web application to support real-time mission planning and decision-making.

APPENDIX

- **Tools and Technologies**
- **Programming Language:** Python
- **Data Collection:** SpaceX REST API, Wikipedia Web Scraping
- **Data Processing:** Pandas, NumPy
- **Data Visualisation:** Matplotlib, Seaborn, Plotly Express
- **Interactive Mapping:** Folium
- **Dashboard Development:** Plotly Dash
- **Machine Learning:** Scikit-learn
- **Database Queries:** SQLite
- **Development Environment:** Jupyter Notebook
- sqlite3
- **Data Sources**
- **SpaceX REST API:** <https://api.spacexdata.com>
- **Wikipedia Falcon 9 Launch Records:**
https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



APPENDIX

Machine Learning Models Evaluated

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)

Key Python Libraries

- pandas
- numpy
- matplotlib
- seaborn
- plotly
- folium
- scikit-learn
- requests
- BeautifulSoup4
- sqlite3

APPENDIX



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GITHUB repository

- **GITHUB repository URL:**

https://github.com/EliRoux/Applied-data-science-Capstone/tree/main/IBM_Applied_Data_Science_Capstone

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- **GITHUB repository Name:**
- **IBM_Applied_data_Science_Capstone**

This repository contains all notebooks, Python scripts, datasets, the Dash application, and the final presentation.

